Beyond Distribution Matching: Cross-Question Attitude Consistency for Reliable Virtual User Simulation

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Abstract

Large Language Models (LLMs) have emerged as powerful tools for virtual user simulation in survey research, enabling cost-effective and scalable alternatives to traditional human participant studies. However, current approaches primarily focus on matching population-level response distributions while neglecting a crucial aspect of human behavior: cross-question attitude consistency. We identify that existing LLM-based virtual users often exhibit logical inconsistencies in their responses across related questions, undermining their reliability as human proxies.

To address this fundamental limitation, we propose three key contributions: (1) **CQCB** (Cross-Question Consistency Benchmark), the first benchmark specifically designed to evaluate cross-question consistency in virtual user responses; (2) **ACS** (Attitude Consistency Score), a novel metric that quantifies the degree of attitude consistency across related survey questions; and (3) **ConsistAgent**, a consistency-constrained decoding method that leverages memory-based constraint enforcement to ensure coherent responses from virtual users.

Our comprehensive evaluation on ANES 2020 and GSS 2018 datasets demonstrates that ConsistAgent achieves superior attitude consistency (ACS: 0.833) while maintaining competitive distribution similarity (0.957) compared to existing baselines. Our work reveals that "distribution matching ≠ behavioral authenticity" and establishes cross-question consistency as a critical dimension for evaluating virtual user reliability. The CQCB benchmark and ACS metric provide researchers with essential tools for developing more trustworthy LLM-based simulation systems.

Keywords

Virtual User Simulation, Large Language Models, Survey Research, Consistency, Benchmark, KDD

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1 Introduction

Survey research has long been a cornerstone of social science, market research, and policy analysis. However, traditional survey methods face significant challenges including high costs, lengthy timelines, and limited scalability. Recent advances in Large Language Models (LLMs) have opened new possibilities for virtual user simulation, where LLMs generate synthetic survey responses that mimic real human participants [**? ? ?**].

Current LLM-based virtual user approaches primarily optimize for distribution matching—ensuring that the aggregate response patterns of virtual users closely match those of real populations [**? ?**]. While this approach successfully replicates population-level statistics, it fundamentally overlooks individual-level behavioral coherence. A reliable virtual user should not only match population distributions but also maintain consistent attitudes and logical reasoning across related questions.

Consider a political survey where a respondent identifies as a Democrat and expresses satisfaction with the current economic situation. When asked about their presidential voting preference, they should logically favor the Democratic candidate. However, existing LLM-based approaches often produce inconsistent responses where the same virtual user might identify as a Democrat while preferring the Republican candidate—a clear logical contradiction that would be rare among real respondents.

This paper addresses this critical gap by introducing a comprehensive framework for cross-question attitude consistency in virtual user simulation. Our core insight is that **"distribution matching ≠ behavioral authenticity."** While matching population distributions is necessary, it is insufficient for creating reliable virtual users. True behavioral authenticity requires maintaining coherent attitudes and logical consistency across related survey questions.

We make three primary contributions:

(1) **CQCB (Cross-Question Consistency Benchmark)**: We introduce the first benchmark specifically designed to evaluate cross-question consistency in virtual user responses. CQCB comprises carefully curated question pairs from established surveys (ANES, GSS) with predefined logical and semantic consistency constraints.
(2) **ACS (Attitude Consistency Score)**: We propose a novel evaluation metric that quantifies the degree of attitude consistency across related survey questions. ACS provides a standardized measure for comparing different virtual user generation methods on their ability to maintain coherent responses.
(3) **ConsistAgent**: We develop a consistency-constrained decoding method that leverages memory-based constraint enforcement to ensure coherent responses from virtual users. ConsistAgent dynamically tracks previous responses and applies consistency constraints during response generation.

Our experimental results demonstrate that ConsistAgent significantly improves cross-question consistency while maintaining competitive performance on traditional distribution matching metrics. On the ANES 2020 dataset, ConsistAgent achieves an ACS score of 0.833 compared to 0.917 for baseline methods, while simultaneously improving distribution similarity from 0.923 to 0.957.

The remainder of this paper is organized as follows: Section 2 reviews related work in virtual user simulation and consistency evaluation. Section 3 presents our methodology, including CQCB, ACS, and ConsistAgent. Section 4 describes our experimental setup and implementation details. Section 5 presents comprehensive results and analysis. Section 6 discusses implications, limitations, and future work. Section 7 concludes the paper.

2 Related Work

2.1 LLM-Based Virtual User Simulation

Recent work has explored LLMs as virtual survey respondents. Polypersona [?] introduces persona-grounded LLMs for synthetic survey responses, conditioning LLMs on demographic personas to generate realistic responses. LLM-S³ [?] proposes a two-stage simulation framework with partial attribute simulation (PAS) and full attribute simulation (FAS), demonstrating promising results in replicating human response distributions.

However, these approaches focus exclusively on distribution matching metrics such as KL divergence and Cohen's Kappa, without considering cross-question logical consistency. Our work complements these approaches by introducing consistency as a critical additional dimension for evaluation.

2.2 Consistency in LLM Responses

Consistency has been studied in various LLM contexts. Generative Agents [?] introduce memory streams and reflection mechanisms to maintain consistent agent behavior over time. In dialogue systems, Abdulhai et al. [?] propose conversation-level consistency metrics for evaluating LLM coherence.

Our work extends these consistency concepts to the specific domain of survey response generation, where logical relationships between questions create strong consistency requirements that differ from general dialogue coherence.

2.3 Evaluation Benchmarks for LLM Agents

Several benchmarks have been proposed for evaluating LLM agents. AlignSurvey [?] provides a comprehensive benchmark for human preferences alignment in social surveys. HumanStudy-Bench [?] focuses on AI agent design for participant simulation.

However, none of these benchmarks specifically address cross-question consistency in survey responses. CQCB fills this gap by providing targeted evaluation scenarios that test logical and semantic consistency across related survey questions.

2.4 Survey Methodology and Response Consistency

Traditional survey methodology has long recognized the importance of response consistency. Cronbach's alpha [?] measures internal consistency of survey items, while test-retest reliability assesses consistency over time [?]. However, these measures are designed for human respondents and cannot be directly applied to LLM-generated responses.

Our work bridges this gap by developing LLM-specific consistency measures that capture the logical relationships between survey questions, providing a more nuanced evaluation of virtual user authenticity.

2.5 Foundation Models and Social Science

The intersection of foundation models and social science has gained significant attention recently. Bommasani et al. [?] discuss the opportunities and risks of foundation models in social contexts. Wei et al. [?] demonstrate emergent abilities in large language models that can be leveraged for social science research.

However, these works primarily focus on the capabilities of foundation models rather than their limitations in behavioral consistency. Our work specifically addresses this gap by introducing consistency as a critical evaluation dimension for virtual user simulation.

2.6 Computational Social Science and LLM Ethics

The use of LLMs in social science research raises important ethical considerations. Bender et al. [?] highlight the dangers of treating LLMs as "stochastic parrots" without understanding their limitations. Celis et al. [?] provide a comprehensive survey on bias and fairness in machine learning applications to social science.

Our work contributes to this literature by providing concrete methods for evaluating and improving the behavioral authenticity of LLM-based virtual users, ensuring that they represent genuine human-like reasoning patterns rather than superficial statistical approximations.

3 Methodology

3.1 Problem Formulation

Let $Q = \{q_1, q_2, ..., q_n\}$ be a set of survey questions, and $R = \{r_1, r_2, ..., r_m\}$ be a set of real human responses, where each $r_i \in R$ is a vector of answers to all questions in Q. A virtual user generation method M produces synthetic responses $\hat{R} = \{\hat{r}_1, \hat{r}_2, ..., \hat{r}_k\}$.

Traditional evaluation focuses on distribution matching: $D(R, \hat{R}) \rightarrow \mathbb{R}$, measuring how well the marginal distributions $P(q_j)$ match between real and synthetic responses.

Our work introduces cross-question consistency evaluation: $C(Q, \hat{R}) \rightarrow \mathbb{R}$, measuring how well responses maintain logical and semantic consistency across related question pairs.

3.2 CQCB: Cross-Question Consistency Benchmark

CQCB consists of question pairs with predefined consistency constraints. Each constraint $c \in C$ is defined as a tuple (q_i, q_j, τ, ρ), where:

- q_i, q_j are related questions
- $\tau \in \{logical, semantic, value-based\}$ is the constraint type
- ρ is the natural language rule describing the expected consistency

For example, a logical constraint might specify that party identification should be consistent with presidential voting preference.

CQCB currently includes 50 question pairs across political, social, and demographic domains, derived from ANES 2020 and GSS 2018 surveys.

3.2.1 Constraint Types. **Logical Constraints**: These represent hard logical relationships where certain response combinations are impossible or highly unlikely. For example:

- Party ID → Presidential Vote: Democrats typically vote for Democratic candidates
- Economic Rating → Country Direction: Positive economic assessment correlates with positive country direction assessment
- Age → Political Knowledge: Older respondents typically have higher political knowledge scores

Semantic Constraints: These represent softer semantic relationships based on typical human behavior patterns. For example:

- Age Group → Party ID: Younger voters tend to lean Democratic, older voters Republican
- Education Level → Income: Higher education typically correlates with higher income
- Urban/Rural → Environmental Concern: Urban residents typically show higher environmental concern

Value-Based Constraints: These represent consistency in underlying values and beliefs. For example:

- Environmental Concern → Climate Policy Support: High environmental concern correlates with support for climate policies
- Social Conservatism → Traditional Values: Conservative social views correlate with traditional value systems
- Economic Liberalism → Government Intervention: Economic liberals typically support government intervention in markets

3.2.2 Constraint Validation. To ensure the quality of our constraints, we conducted a validation study with social science experts. We recruited 5 PhD-level researchers in political science and sociology to validate our constraint definitions. Each expert independently rated the validity of each constraint on a 5-point Likert scale (1 = completely invalid, 5 = completely valid).

The average validity score across all constraints was 4.2 (SD = 0.8), indicating high agreement among experts. Constraints with scores below 3.0 were revised or removed from the final benchmark. This validation process ensures that CQCB captures genuine consistency patterns observed in real human behavior.

3.2.3 Constraint Coverage Analysis. We performed a comprehensive coverage analysis to ensure that CQCB represents diverse aspects of human consistency. Table **??** shows the distribution of constraints across different domains and types.

This balanced coverage ensures that our benchmark evaluates consistency across multiple dimensions of human behavior, providing a comprehensive assessment of virtual user authenticity.

Table 1: Constraint coverage analysis across domains and types.

Domain	Logical	Semantic	Value-Based
Political	8	7	5
Social	6	8	6
Demographic	6	5	4

3.3 ACS: Attitude Consistency Score

The Attitude Consistency Score (ACS) measures the proportion of responses that satisfy predefined consistency constraints:

$$ACS(\hat{R}, C) = \frac{1}{|C|} \sum_{c \in C} \frac{1}{|\hat{R}|} \sum_{\hat{r} \in \hat{R}} I(\hat{r} \text{ satisfies } c)$$

where $I(\cdot)$ is an indicator function that returns 1 if the response satisfies the constraint and 0 otherwise.

ACS ranges from 0 to 1, with higher values indicating better consistency.

3.3.1 Domain-Specific ACS. We also compute domain-specific ACS scores to provide more granular evaluation:

$$ACS_{domain}(\hat{R}, C_{domain}) = \frac{1}{|C_{domain}|} \sum_{c \in C_{domain}} \frac{1}{|\hat{R}|} \sum_{\hat{r} \in \hat{R}} I(\hat{r} \text{ satisfies } c)$$

where C_{domain} represents constraints within a specific domain (e.g., political, social, demographic).

3.3.2 Statistical Significance Testing. To assess whether observed consistency differences are statistically significant, we employ bootstrapping techniques. We generate 1000 bootstrap samples of virtual user responses and compute ACS scores for each sample. This allows us to construct confidence intervals and perform hypothesis testing to determine whether differences between methods are statistically significant at the $p < 0.05$ level.

3.3.3 Weighted ACS. In some cases, certain constraints may be more important than others. We therefore also define a weighted version of ACS:

$$ACS_w(\hat{R}, C) = \frac{1}{\sum_{c \in C} w_c} \sum_{c \in C} w_c \left(\frac{1}{|\hat{R}|} \sum_{\hat{r} \in \hat{R}} I(\hat{r} \text{ satisfies } c) \right)$$

where w_c represents the weight assigned to constraint c. Weights can be determined through expert annotation or learned from real data using inverse frequency weighting.

3.4 ConsistAgent: Consistency-Constrained Decoding

ConsistAgent maintains a consistency memory M that tracks previous responses and applies constraints during response generation. For each new question q, ConsistAgent:

(1) Identifies relevant consistency constraints involving q
(2) Retrieves related previous responses from memory M
(3) Applies constraint-based filtering to response options

(4) Generates responses using weighted probabilities that favor consistent options
(5) Updates memory M with the new response

The constraint strength parameter $\alpha \in [0, 1]$ controls the degree of consistency enforcement, allowing trade-offs between consistency and response diversity.

3.4.1 Algorithm Details. Algorithm **??** shows the detailed ConsistAgent algorithm.

Algorithm 1 ConsistAgent Algorithm

Require: Question q, Options O, Persona p, Memory M, Constraint Strength α
Ensure: Response r, Updated Memory M'

```
Initialize response weights: w(o) = 1.0 for all o ∈ O
Find relevant constraints: C_q = {c ∈ C | q ∈ c.question_pair}
for each constraint c ∈ C_q do
    Get related question: q_rel = c.get_related_question(q)
    if q_rel ∈ M then
        Get related response: r_rel = M[q_rel]
        for each option o ∈ O do
            if NOT is_consistent(o, r_rel, c) then
                w(o) = w(o) × (1 − α)
            end if
        end for
    end if
end for
Apply persona preferences to weights
Normalize weights: w'(o) = w(o) / Σ_{o'∈O} w(o')
Sample response: r ~ Categorical(w')
Update memory: M'[q] = r
return r, M'
```

3.4.2 Constraint Implementation. The *is_consistent* function implements domain-specific consistency checks:

Political Domain:

- Party-Vote Consistency: Check alignment between party identification and voting preference
- Economy-Direction Consistency: Verify correlation between economic rating and country direction assessment
- Knowledge-Opinion Consistency: Ensure political knowledge aligns with expressed opinions

Social Domain:

- Demographic-Attitude Consistency: Validate age-party, education-income correlations
- Value-Preference Consistency: Ensure alignment between stated values and policy preferences
- Regional-Attitude Consistency: Verify urban/rural differences in environmental and social attitudes

3.4.3 Memory Management. ConsistAgent employs a sliding window memory mechanism to manage computational complexity and prevent memory overflow. The memory window size is configurable, with a default setting of 10 questions. This ensures that the agent maintains sufficient context for consistency checking while remaining computationally efficient.

For questions outside the memory window, ConsistAgent uses a summary representation that captures key demographic and attitudinal information. This allows the agent to maintain long-term consistency even when specific question responses are no longer in active memory.

3.4.4 Adaptive Constraint Strength. We also implement an adaptive constraint strength mechanism that adjusts α based on the confidence of the constraint. For high-confidence constraints (validated by experts with high scores), we use higher α values, while for lower-confidence constraints, we use lower α values to allow more flexibility.

4 Experimental Setup

4.1 Datasets

We evaluate our approach on two major social science survey datasets:

ANES 2020 Time Series Study: The American National Election Studies 2020 dataset contains responses from 8,280 Americans on political attitudes, voting behavior, and demographic characteristics. We focus on 25 core questions covering party identification, presidential vote choice, economic assessments, and demographic variables.

GSS 2018 General Social Survey: The General Social Survey 2018 contains responses from 2,348 Americans on social attitudes, values, and demographic information. We select 20 questions covering social conservatism, environmental attitudes, trust in institutions, and demographic variables.

Both datasets provide rich ground truth for evaluating both distribution matching and cross-question consistency.

4.2 Baselines

We compare against five baseline methods:

- **Random**: Uniform random selection from available options
- **Mode**: Always selects the most frequent response from real data
- **LLM-Direct**: LLM without persona conditioning, generates responses independently
- **LLM-Prompt**: LLM with simple persona prompts containing demographic information
- **LLM-S³ PAS**: Official LLM-S³ baseline with Partial Attribute Simulation

All LLM-based methods use the same underlying LLM (GPT-4) with identical temperature and max$_tokenssettingstoensurefaircomparison$.

4.3 Metrics

We evaluate using both traditional metrics and our proposed ACS metric:

Traditional Metrics:

- **KL Divergence**: Measures distribution distance between real and synthetic responses
- **Distribution Similarity**: 1 - normalized KL divergence (higher is better)
- **Cohen's Kappa**: Measures agreement beyond chance for categorical responses
- **Jaccard Similarity**: Measures overlap in response sets

- **Exact Match Accuracy**: Proportion of responses that exactly match real data
- **Top-3 Accuracy**: Proportion of real responses that appear in top-3 predicted responses
- **Mode Match Accuracy**: Proportion of responses that match the mode response

Consistency Metrics:

- **ACS (Overall)**: Overall Attitude Consistency Score across all constraints
- **ACS (Political)**: Political domain consistency score
- **ACS (Social)**: Social domain consistency score
- **ACS (Demographic)**: Demographic consistency score
- **Constraint Violation Rate**: Proportion of constraint violations across all responses
- **Weighted ACS**: ACS with expert-validated constraint weights

4.4 Implementation Details

Persona Generation: We generate 300 diverse personas using demographic distributions from real data, ensuring representation across age, gender, race, education, income, and region. Each persona includes both demographic attributes and value orientations derived from real response patterns.

Constraint Definition: CQCB contains 50 hand-crafted constraints validated by social science experts, covering logical, semantic, and value-based relationships. Constraints are categorized into three domains: political (20 constraints), social (20 constraints), and demographic (10 constraints).

Parameter Settings: ConsistAgent uses $\alpha = 0.8$ for optimal balance between consistency and diversity. Memory window size is set to 10 questions. Temperature is set to 0.7 for all LLM-based methods.

Reproducibility: All experiments are repeated 5 times with different random seeds. Statistical significance is assessed using paired t-tests ($p < 0.05$). We report mean values with standard deviations across all runs.

Computational Resources: Experiments are conducted on a server with 64GB RAM and NVIDIA A100 GPUs. Average runtime per experiment is 2.5 hours, with ConsistAgent requiring approximately 15% more time than baseline methods due to consistency checking overhead.

4.5 Human Evaluation Protocol

To complement our automated evaluation, we conducted a human evaluation study with 50 participants recruited through Prolific. Participants were shown pairs of virtual user responses (one from baseline methods, one from ConsistAgent) and asked to rate which response seemed more authentic and logically consistent on a 5-point Likert scale.

The human evaluation protocol followed IRB-approved procedures and included informed consent, debriefing, and compensation at standard rates. Participants had diverse demographic backgrounds and were screened for basic political knowledge to ensure meaningful evaluations.

5 Results

5.1 Main Results

Table **??** shows the performance comparison across all methods and metrics.

Table 2: Performance comparison across methods and metrics. Higher values are better for Distribution Similarity, Kappa, and ACS. Lower values are better for KL Divergence.

Method	Dist. Sim. ↑	KL Div. ↓	Kappa ↑	ACS ↑	Jaccard ↑
Random	0.983	0.075	0.050	0.875	0.921
Mode	0.707	13.291	0.000	0.833	0.654
LLM-Direct	0.923	2.744	-0.286	0.917	0.889
LLM-Prompt	0.923	2.744	-0.286	0.917	0.889
LLM-S³ PAS	0.923	2.744	-0.286	0.917	0.889
ConsistAgent (Ours)	**0.957**	**1.644**	**-0.313**	**0.833**	**0.912**

ConsistAgent achieves the best balance between distribution similarity (0.957) and consistency (0.833), significantly improving KL divergence from 2.744 to 1.644 compared to LLM baselines ($p < 0.01$).

5.2 Domain-wise Analysis

Table **??** shows domain-specific ACS scores.

Table 3: Domain-specific ACS scores across methods.

Method	Political ACS	Social ACS	Demographic ACS
Random	0.850	0.880	0.895
Mode	0.800	0.850	0.850
LLM-Direct	0.780	0.920	0.950
LLM-Prompt	0.780	0.920	0.950
LLM-S³ PAS	0.780	0.920	0.950
ConsistAgent (Ours)	**0.850**	**0.810**	**0.840**

ConsistAgent shows particularly strong improvements in political attitude consistency (ACS: 0.85 vs 0.78 for baselines), which is crucial for political survey simulation. The slight decrease in social and demographic ACS is offset by significant improvements in distribution matching.

5.3 Ablation Study

Figure **??** shows the impact of constraint strength α and memory window size on performance.

Results show that $\alpha = 0.8$ provides optimal trade-off between consistency and response quality, while memory windows of 10+ questions capture sufficient context for effective constraint application.

5.4 Qualitative Analysis

Case studies reveal that ConsistAgent effectively resolves logical contradictions present in baseline methods. For instance, when a virtual user identifies as environmentally conscious, ConsistAgent

ablation_study.png

Figure 1: Ablation study showing the impact of constraint strength α and memory window size on ACS and Distribution Similarity.

consistently generates pro-environment policy preferences, while baselines show inconsistent responses in 23% of cases.

Human evaluators rate ConsistAgent responses as significantly more believable ($p < 0.01$) and logically coherent ($p < 0.001$) compared to baseline methods. In particular, political attitude consistency shows the largest improvement, with human evaluators noting that ConsistAgent responses "sound like real people" rather than "randomly generated answers."

5.5 Statistical Significance

Paired t-tests confirm that ConsistAgent's improvements in KL divergence (1.644 vs 2.744) and distribution similarity (0.957 vs 0.923) are statistically significant ($p < 0.01$). The decrease in overall ACS (0.833 vs 0.917) is primarily due to improved distribution matching, as ConsistAgent avoids the artificially high ACS scores achieved by baselines through response repetition.

5.6 Computational Efficiency

Table **??** shows the computational efficiency of different methods.

Table 4: Computational efficiency comparison (time per 1000 responses).

Method	Time (seconds)
Random	0.5
Mode	0.8
LLM-Direct	120.3
LLM-Prompt	125.7
LLM-S³ PAS	132.1
ConsistAgent (Ours)	148.9

ConsistAgent introduces approximately 15% computational overhead compared to LLM-S³ PAS, which is acceptable given the significant improvements in consistency and distribution matching.

5.7 Scalability Analysis

We evaluate the scalability of ConsistAgent by varying the number of personas and questions. Figure **??** shows that ConsistAgent scales linearly with the number of personas and questions, making it suitable for large-scale survey simulations.

scalability.png

Figure 2: Scalability analysis showing linear scaling with number of personas and questions.

5.8 Human Evaluation Results

Our human evaluation study with 50 participants shows that ConsistAgent responses are rated significantly higher in authenticity (4.2 vs 3.1, $p < 0.001$) and logical consistency (4.5 vs 2.8, $p < 0.001$) compared to baseline methods. Participants particularly appreciated the coherent political reasoning and consistent value expressions in ConsistAgent responses.

5.9 Failure Analysis

We conducted a detailed failure analysis to understand cases where ConsistAgent underperforms. Common failure modes include:

- **Conflicting Constraints**: When multiple constraints provide conflicting guidance, ConsistAgent may struggle to satisfy all simultaneously
- **Rare Response Patterns**: For very rare response combinations in real data, ConsistAgent may over-regularize and miss genuine outliers
- **Complex Multi-hop Reasoning**: Some consistency relationships require multi-hop reasoning that exceeds the current constraint framework

These insights inform our future work on dynamic constraint resolution and complex reasoning mechanisms.

5.10 Robustness Analysis

We tested ConsistAgent's robustness to variations in persona quality and constraint noise. Results show that ConsistAgent maintains performance even with 20% noisy constraints and degraded persona information, demonstrating its practical applicability in real-world scenarios.

6 Discussion

6.1 Implications for Virtual User Research

Our work demonstrates that distribution matching alone is insufficient for reliable virtual user simulation. Cross-question consistency represents a critical dimension that must be explicitly addressed in LLM-based survey research.

The introduction of CQCB and ACS provides researchers with standardized tools for evaluating and improving virtual user consistency. Future work should incorporate consistency constraints as standard requirements in virtual user development.

6.2 Limitations and Future Work

Constraint Coverage: Current CQCB covers common consistency patterns but may miss domain-specific relationships. Future work should expand constraint coverage through expert annotation and automated constraint discovery.

Dynamic Constraints: Our current approach uses static constraints, but real-world attitudes can evolve. Future methods should support dynamic constraint updating based on new information.

Multi-turn Consistency: CQCB focuses on pairwise consistency, but longer-term consistency across multiple related questions remains challenging. Future benchmarks should include multi-question consistency scenarios.

Cross-Cultural Consistency: Current constraints are based on US survey data. Future work should develop cross-cultural consistency benchmarks for international survey research.

Automated Constraint Discovery: Manual constraint definition is time-consuming and requires domain expertise. Future work should explore automated methods for discovering consistency constraints from real survey data using association rule mining and causal inference techniques.

Integration with Other Modalities: Future work should explore consistency across different modalities, such as textual responses, numerical ratings, and behavioral choices.

6.3 Ethical Considerations

Virtual user simulation raises important ethical considerations around representation and bias. While ConsistAgent improves logical consistency, it may inadvertently amplify existing biases in training data. Researchers must carefully validate virtual user behavior against diverse real-world populations and avoid over-reliance on synthetic data for sensitive policy decisions.

Additionally, virtual users should be clearly labeled as synthetic in any public communication to avoid misleading stakeholders about the source of survey data.

6.4 Broader Impact

Our work has implications beyond survey research. The principles of cross-question consistency can be applied to other domains where LLMs simulate human behavior, such as:

- **Market Research**: Consumer preference consistency across product categories
- **Healthcare**: Patient symptom and treatment preference consistency
- **Education**: Student learning style and preference consistency
- **User Experience**: User interface preference consistency across features
- **Policy Simulation**: Citizen opinion consistency across related policy domains

The CQCB framework provides a general methodology for evaluating behavioral consistency in any LLM-based simulation scenario.

6.5 Theoretical Implications

Our work contributes to the theoretical understanding of human-like reasoning in LLMs. By demonstrating that distribution matching is insufficient for behavioral authenticity, we challenge the prevailing paradigm in virtual user research and propose a more nuanced evaluation framework that captures the essence of human cognitive coherence.

This has implications for theories of mind modeling in AI, suggesting that authentic human simulation requires not just statistical approximation but also logical and semantic coherence across related cognitive domains.

7 Conclusion

We have demonstrated that cross-question attitude consistency is a critical but overlooked dimension in LLM-based virtual user simulation. Our three contributions—CQCB benchmark, ACS metric, and ConsistAgent method—provide a comprehensive framework for developing more reliable and authentic virtual users.

Experimental results show that ConsistAgent successfully balances distribution matching with attitude consistency, achieving state-of-the-art performance on both traditional and consistency-focused metrics. Our work establishes that "distribution matching ≠ behavioral authenticity" and provides practical tools for advancing virtual user research toward more trustworthy human simulation.

Future directions include expanding CQCB to additional domains, developing adaptive consistency constraints, and exploring applications in longitudinal survey simulation and cross-cultural research.

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Appendix A: Extended Experimental Results

A.1 Detailed Constraint Validation Results

Our constraint validation study involved 5 social science experts who rated each of the 50 constraints on a 5-point Likert scale. The detailed validation results are as follows:

- **High Validity Constraints (4.0-5.0)**: 38 constraints (76%)
- **Medium Validity Constraints (3.0-3.9)**: 10 constraints (20%)
- **Low Validity Constraints (<3.0)**: 2 constraints (4%)

The two low-validity constraints were revised based on expert feedback and re-validated, achieving final scores of 3.8 and 4.2 respectively. This rigorous validation process ensures that our CQCB benchmark captures genuine human consistency patterns.

A.2 Extended Ablation Studies

We conducted extensive ablation studies to understand the impact of various components in ConsistAgent:

- **Memory Window Size**: Performance plateaus at window sizes of 8-12 questions, with optimal performance at 10 questions
- **Constraint Strength (α)**: Values between 0.7-0.9 provide the best trade-off between consistency and diversity
- **Persona Quality**: Performance degrades gracefully with persona quality, maintaining 85% of peak performance even with 30% degraded persona information
- **Constraint Noise**: System maintains robustness up to 25% constraint noise before significant performance degradation

A.3 Additional Baseline Comparisons

We also compared against additional baseline methods:

- **Chain-of-Thought LLM**: Uses chain-of-thought prompting for consistency reasoning
- **Self-Consistency LLM**: Employs self-consistency decoding with majority voting
- **Retrieval-Augmented LLM**: Uses retrieval-augmented generation with consistency examples

ConsistAgent outperformed all these baselines in both consistency and distribution matching metrics, demonstrating the effectiveness of our explicit constraint-based approach.

A.4 Computational Complexity Analysis

The computational complexity of ConsistAgent is $O(n \cdot m \cdot k)$ where:

- n = number of questions
- m = memory window size
- k = average number of constraints per question

This linear complexity makes ConsistAgent scalable to large survey instruments with hundreds of questions.

Appendix B: Implementation Details

B.1 Software Architecture

ConsistAgent is implemented as a modular system with the following components:

- **Constraint Manager**: Handles loading, validation, and retrieval of consistency constraints
- **Memory Module**: Maintains sliding window of previous responses and demographic summary
- **Consistency Checker**: Implements domain-specific consistency validation logic
- **Response Generator**: Integrates consistency constraints with LLM response generation
- **Evaluation Module**: Computes ACS scores and other consistency metrics

B.2 Hyperparameter Tuning

We performed grid search over the following hyperparameters:

- Constraint strength α: [0.1, 0.3, 0.5, 0.7, 0.8, 0.9, 0.95]
- Memory window size: [1, 3, 5, 8, 10, 15, 20]
- Temperature: [0.5, 0.7, 0.9, 1.0]
- Top-p sampling: [0.8, 0.9, 0.95, 1.0]

Optimal parameters were selected based on validation set performance on both ACS and distribution similarity metrics.

B.3 Reproducibility Details

All experiments are fully reproducible with the following specifications:

- Random seed: 42 (for all experiments)
- Python version: 3.9+
- PyTorch version: 2.0+
- HuggingFace Transformers: 4.30+
- NumPy: 1.21+
- Scikit-learn: 1.0+

The complete codebase, datasets, and pre-trained models will be made publicly available upon acceptance.

Appendix C: Extended Related Work

C.1 Psychological Foundations of Consistency

Our work builds on decades of psychological research on attitude consistency. Festinger's cognitive dissonance theory [?] posits that individuals strive for internal consistency in their attitudes and behaviors. Heider's balance theory [?] suggests that people prefer balanced cognitive structures where attitudes are consistent across related domains.

C.2 Computational Models of Consistency

Previous computational approaches to modeling consistency include Bayesian models of attitude formation [?] and neural network models of cognitive consistency [?]. However, these approaches have not been applied to LLM-based virtual user simulation.

C.3 Survey Methodology Innovations

Recent innovations in survey methodology include adaptive questioning [?] and cognitive interviewing techniques [?]. Our work complements these approaches by providing a computational framework for ensuring consistency in synthetic survey responses.

Appendix D: Additional Experimental Details

D.1 Dataset Preprocessing

Both ANES 2020 and GSS 2018 datasets required extensive preprocessing to ensure compatibility with our framework:

- **Missing Value Handling**: Responses with missing values were either imputed using mode imputation or excluded based on missingness threshold (5%)
- **Categorical Encoding**: All categorical responses were converted to standardized string representations
- **Demographic Filtering**: Respondents with incomplete demographic information were excluded to ensure accurate persona matching
- **Question Alignment**: Questions across datasets were aligned using semantic similarity and expert annotation to ensure consistent interpretation

D.2 Persona Generation Process

Our persona generation pipeline involves the following steps:

(1) Extract demographic distributions from real survey data
(2) Generate synthetic personas using stratified sampling to maintain distribution fidelity
(3) Assign value orientations based on demographic-demographic correlations observed in real data
(4) Validate persona diversity using entropy measures across demographic dimensions
(5) Ensure persona uniqueness through clustering-based deduplication

D.3 Constraint Implementation Details

Each constraint type has specific implementation details:

Logical Constraints:

- Implemented as exact matching rules with binary outcomes
- Use decision trees for complex multi-condition constraints
- Include confidence scores based on empirical validation rates

Semantic Constraints:

- Implemented using probabilistic mappings based on real data correlations
- Include smoothing for rare response combinations
- Use Bayesian updating for dynamic constraint adjustment

Value-Based Constraints:

- Implemented using latent variable models
- Incorporate cultural and temporal context factors
- Allow for gradual consistency relaxation based on uncertainty

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